CS182: Introduction to Machine Learning, Fall 2025
Homework 3

(Due on Wednesday, Dec. 10 at 11:59 pm (CST))

Notice:

- Please submit your assignments via Gradescope. The entry code is 428YBY.
- Please make sure you select your answer to the corresponding question when submitting your assignments.
- Each person has a total of five days to be late without penalty for all the assignments. Each late delivery less than one day will be counted as one day.

1. [20 points] [*Multilayer Perceptrons*]You are designing a decoupled controller for a 2-DOF (Degree of Freedom) robotic arm. The network takes the joint angles (θ_1, θ_2) as input and simultaneously outputs the required torques (τ_1, τ_2) for both joints.

   Architecture:

   - Input Layer: 2 neurons ($\mathbf{x} \in \mathbb{R}^2$).
   - Hidden Layer: 2 neurons with *ReLU* activation: $\mathbf{h} = \text{ReLU}(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$.
   - Output Layer: 2 neurons with *Linear* activation: $\hat{\mathbf{y}} = \mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)}$.

   Parameters: The weights and biases are initialized as follows:

$$\mathbf{W}^{(1)} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{b}^{(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{W}^{(2)} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{b}^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

   (a) Given the joint configuration input $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, compute the predicted torque vector $\hat{\mathbf{y}}$. [5 points]

   (b) The optimal torque provided by the physics engine (ground truth) is $\mathbf{y} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$.

   We define the loss as the Mean Squared Error (scaled):

$$L = \frac{1}{2}\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \frac{1}{2}\sum_i (y_i - \hat{y}_i)^2$$

   Using the prediction $\hat{\mathbf{y}}$, compute the gradients of the loss with respect to the first-layer weights $\mathbf{W}^{(1)}$.[10 points]

   (c) Another engineer proposes changing the output activation to a Sigmoid function ($f(z) = \frac{1}{1+e^{-z}}$) to make the output "smoother". Is using a Sigmoid output appropriate for this regression task? Explain why or why not.[5 points]

2. [20 points] [*Support Vector Machines*]

(a) Given the positive class points $x_1 = (1, 2)^T, \quad x_2 = (2, 3)^T, \quad x_3 = (3, 3)^T$, and the negative class points $x_4 = (2, 1)^T, \quad x_5 = (3, 2)^T$, find the maximum–margin separating hyperplane and the classification decision function. Then, plot the separating hyperplane, the margin boundaries, and the support vectors on a 2D plane. (using sklearn, and provide both the code and the generated results.) [10 points]

(b) A linear support vector machine can also be defined as follows:

$$\min_{w,b,\zeta} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \zeta_i^2$$

subject to

$$y_i(w \cdot x_i + b) \geq 1 - \zeta_i, \quad \forall i = 1, 2, \ldots, N$$

$$\zeta_i \geq 0, \quad \forall i = 1, 2, \ldots, N$$

Find its dual formulation, and please provide detailed derivations. [10 points]

3. [20 points] [*Kernel Extension*]

In this part, let us try to get a better intuition about kernels themselves.

Definition 1 (Positive definite kernel)**.** *A function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a positive definite kernel iff it is symmetric, that is, $k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}', \mathbf{x})$ for any two objects $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, and positive definite definite, that is,*

$$\sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0$$

for any $n \geq 0$, any choice of n objects $x_1 \cdots, x_n \in \mathcal{X}$, and any choice of real numbers $c_1, \cdots, c_n \in \mathbb{R}$.

From now on, we only focus on positive definite kernels, and simply call them *kernels*.

Kernels as Inner Product For general objects $\mathbf{x} \in \mathcal{X}$, we first represent each object $\mathbf{x} \in \mathcal{X}$ as a vector $\phi(\mathbf{x}) \in \mathbb{R}^p$, and then defining a kernel for any $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ by

$$k(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$$

Kernels as Measures of Similarity Kernels are often presented as measures of similarity, in the sense that $k(\mathbf{x}, \mathbf{x}')$ is "large" when $\mathbf{x}$ and $\mathbf{x}'$ are "similar".

(a) Consider the following kernel and give a relevant interpretations as a measure of similarity. You can just describe it in natural language.

$$k(\mathbf{x}, \mathbf{x}') = \exp\{-\frac{d(\mathbf{x}, \mathbf{x}'^2)}{2\sigma^2}\}$$

where σ is a parameter and d is the Euclidean distance.[4 points]

Kernels as Measures of Function Regularity Now we associate kernel with some functional space and norms associated with it. Let k be a kernel on a space $\mathcal{X}$. The set $\mathcal{H}_k$ is defined as the set of function $f : \mathcal{X} \to \mathbb{R}$ of the form

$$f(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x})$$

for $n > 0$, a finite number of points $\mathbf{x}_1, \cdots, \mathbf{x}_n \in \mathcal{X}$, and a finite number of weights $\alpha_1, \cdots, \alpha_n \in \mathbb{R}$, together with

$$||f||^2_{\mathcal{H}_k} := \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j)$$

The dot product defined for two elements $f(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x})$ and $g(\mathbf{x}) = \sum_{j=1}^{m} \alpha'_j k(\mathbf{x}'_j, \mathbf{x})$ is

$$\langle f, g \rangle = \sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_i \alpha'_j k(\mathbf{x}_i, \mathbf{x}'_j)$$

(b) Express the value $f(\mathbf{x})$ of a function $f \in \mathcal{H}_k$ at a point $\mathbf{x} \in \mathcal{X}$ as a dot product in $\mathcal{H}_k$.[3 points]

(c) Write down the corresponding f, g when $\langle f, g \rangle = k(\mathbf{x}, \mathbf{x}')$.[3 points]

Most kernel methods (including SVMs) have an interpretation in terms of functional analysis and can be defined as algorithms that, given a set of objects $\mathcal{S}$, return a function that solves the equation

$$\min_{f \in \mathcal{H}_k} E(f, \mathcal{S}) + g(\|f\|_{\mathcal{H}_k})$$

(d) (The Representer Theorem) Consider a positive-definite real-valued kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ on a non-empty set $\mathcal{X}$ with a corresponding reproducing kernel space $\mathcal{H}_k$. Let there be given

- a training sample $(x_1, y_1), \ldots, (x_n, y_n) \in \mathcal{X} \times \mathbb{R}$
- a strictly increasing real-valued function $g\colon [0, \infty) \to \mathbb{R}$
- an arbitrary error function $E\colon (\mathcal{X} \times \mathbb{R}^2)^n \to \mathbb{R} \cup \{\infty\}$

which together define the following regularized empirical risk functional on $\mathcal{H}_k$

$$f \mapsto E\left((x_1, y_1, f(x_1)), \ldots, (x_n, y_n, f(x_n))\right) + g\left(\|f\|_{\mathcal{H}_k}\right)$$

Show that any minimizer of the empirical risk

$$f^* = \underset{f \in H_k}{\operatorname{argmin}} \left\{E\left((x_1, y_1, f(x_1)), \ldots, (x_n, y_n, f(x_n))\right) + g\left(\|f\|_{\mathcal{H}_k}\right)\right\}, \quad (*)$$

admits a representation of the form

$$f^*(\cdot) = \sum_{i=1}^{n} \alpha_i k(\cdot, x_i)$$

[10 points]

4. [20 points] [*Coding*] Complete the 'Problem4.ipynb'. Please include the code you need to write and the output results (which may include values and images) in the PDF file for submission.

5. [20 points] [*Principal Component Analysis (PCA)*] As discussed in class, Principal Component Analysis (PCA) can be derived from a reconstruction error perspective. Consider the following formulation:

   Given a dataset $\{\mathbf{x}^t\}_{t=1}^N$ where $\mathbf{x}^t \in \mathbb{R}^d$, we aim to represent each data point in a lower-dimensional space and then reconstruct it with minimal error. The reconstruction model is:

$$\hat{\mathbf{x}}^t = \mathbf{m} + \mathbf{W}\mathbf{z}^t$$

   where $\mathbf{m} \in \mathbb{R}^d$ is a common bias vector, $\mathbf{W} \in \mathbb{R}^{d \times k}$ is the projection matrix, and $\mathbf{z}^t \in \mathbb{R}^k$ is the low-dimensional representation of $\mathbf{x}^t$. The optimization problem is to minimize the total reconstruction error:

$$\text{(P-RE)} \quad \underset{\mathbf{m},\mathbf{W},\{\mathbf{z}^t\}}{\text{minimize}} \quad \frac{1}{2}\sum_{t=1}^{N} \left\|\mathbf{x}^t - (\mathbf{m} + \mathbf{W}\mathbf{z}^t)\right\|_2^2$$
$$\text{subject to} \quad \mathbf{W}^T\mathbf{W} = \mathbf{I}_k$$

   Assume we center the data by setting $\mathbf{m}$ to be the sample mean, i.e., $\mathbf{m} = \frac{1}{N}\sum_{t=1}^{N} \mathbf{x}^t$, and define the centered data points as $\tilde{\mathbf{x}}^t = \mathbf{x}^t - \mathbf{m}$.

   (a) [4 points] Show that for a fixed $\mathbf{W}$, the optimal low-dimensional representation $\mathbf{z}^t$ for each data point is given by the projection $\mathbf{z}^t = \mathbf{W}^T\tilde{\mathbf{x}}^t$.

   (b) [8 points] Substitute this optimal $\mathbf{z}^t$ back into the objective function of (P-RE). Show that the problem of finding the optimal $\mathbf{W}$ reduces to:

$$\underset{\mathbf{W}}{\text{maximize}} \quad \sum_{t=1}^{N} \tilde{\mathbf{x}}^{tT}\mathbf{W}\mathbf{W}^T\tilde{\mathbf{x}}^t \quad \text{subject to} \quad \mathbf{W}^T\mathbf{W} = \mathbf{I}_k$$

   (c) [8 points] Further show that this maximization problem is equivalent to finding the k eigenvectors of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N}\sum_{t=1}^{N} \tilde{\mathbf{x}}^t\tilde{\mathbf{x}}^{tT}$ corresponding to its k largest eigenvalues. (**Hint:** Use the property of the cyclic property of the trace $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$.)